

Top-level inputs
h : vector of 8 signed fixed point $\langle 1.8 \rangle$
J : vector of 28 signed fixed point $\langle 1.8 \rangle$
spins : bitstring $\langle 8 \rangle$
mask : bitstring $\langle 8 \rangle$

Inputs	Helper: selected_field	Outputs
h : vector of 8 signed fixed point $\langle 1.8 \rangle$	one-hot multiplexer selecting the field of the flipped spin	h_i : signed fixed point $\langle 1.8 \rangle$
mask : bitstring $\langle 8 \rangle$		

Inputs	Helper: selected_coupling_for_j	Outputs
J : vector of 28 signed fixed point $\langle 1.8 \rangle$	8 parallel one-hot multiplexers using packed upper-triangular indexing	J_{ij} : vector of 8 signed fixed point $\langle 1.8 \rangle$
mask : bitstring $\langle 8 \rangle$		
j : integer, $0, \dots, 7$		

Inputs	Helper: spin_value	Outputs
spins : bitstring $\langle 8 \rangle$	bit-to-Ising conversion: $0 \mapsto -1$, $1 \mapsto +1$	s_j : vector of 8 signed fixed point $\langle 1.8 \rangle$

Inputs	Helper: selected_spin	Outputs
spins : bitstring $\langle 8 \rangle$	bit selection using the one-hot mask	s_i : signed fixed point $\langle 1.8 \rangle$
mask : bitstring $\langle 8 \rangle$		

Inputs	Main block: local field terms	Outputs
h_i : signed fixed point $\langle 1.8 \rangle$	forms local-field inputs: $\text{terms}[0] = h_i$ and $\text{terms}[j+1] = J_{ij}s_j$	terms : vector of 9 signed fixed point $\langle 1.8 \rangle$
J_{ij} : vector of 8 signed fixed point $\langle 1.8 \rangle$		
s_j : vector of 8 signed fixed point $\langle 1.8 \rangle$		

Inputs	Main block: tree_sum $\langle 9 \rangle$	Outputs
terms : vector of 9 signed fixed point $\langle 1.8 \rangle$	fully unrolled binary adder tree	field_over_alpha : signed fixed point $\langle 1.8 \rangle$

Inputs	Main block: local_delta_over_alpha	Outputs
field_over_alpha : signed fixed point $\langle 1.8 \rangle$	applies the selected spin sign and computes $2s_i \text{field}_i / \alpha$; doubling is implemented by addition	delta_over_alpha : signed fixed point $\langle 1.8 \rangle$
s_i : signed fixed point $\langle 1.8 \rangle$		

Inputs	Helper: positive_part	Outputs
delta_over_alpha : signed fixed point $\langle 1.8 \rangle$	Metropolis clipping: $x = \max(\delta, 0)$	x : signed fixed point $\langle 1.8 \rangle$

Inputs	Main block: piecewise cutoff	Outputs
x : signed fixed point $\langle 1.8 \rangle$	returns 1 at $x = 0$, returns 0 at $x \geq x_{\text{cut}}$, otherwise enables the polynomial path	early_prob : unsigned fixed point $\langle 1.31 \rangle$
X_{cut} : signed fixed point $\langle 1.8 \rangle$		poly_enable : boolean

Inputs	Main block: Chebyshev rescaling	Outputs
x : signed fixed point $\langle 1.8 \rangle$	computes $z = x/x_{\text{cut}}$ and then $y = 2z - 1$	y : signed fixed point $\langle 4.36 \rangle$
X_{cut}^{-1} : unsigned fixed point $\langle 16.32 \rangle$		

Inputs	Main block: chebyshev_g / Clenshaw	Outputs
y : signed fixed point $\langle 4.36 \rangle$	evaluates $p(y) = \sum_k c_k T_k(y)$ using offline-generated coefficients	p : signed fixed point $\langle 4.36 \rangle$
c_k : vector of $d+1$ signed fixed point $\langle 4.36 \rangle$		

Top-level outputs
δ_x : pointer to signed fixed point $\langle 1.8 \rangle$
accept_prob : pointer to unsigned fixed point $\langle 1.31 \rangle$

Inputs	Helper: probability clipping	Outputs
p : signed fixed point $\langle 4.36 \rangle$	clips to $[0, 1]$ and selects early or polynomial path	accept_prob : unsigned fixed point $\langle 1.31 \rangle$
early_prob : unsigned fixed point $\langle 1.31 \rangle$		